AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

T103/TM103
Computer Organization and Architecture

Final Examination
Spring Semester 2025/2026

Date: 11/Jun./2026

Number of Exam Pages: 4 (including this cover sheet)	Time Allowed: 2 Hours

Instructions:

- Total Marks: **50**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Programmable Calculators** are **not** allowed.

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Part I: Choose the most correct answer [10 marks]

This part consists of **10 questions** carrying the weight **of ONE mark each**.
Q-1: Answer all the following questions by choosing the most correct statement.

1. Which component is considered the **"brain" of the computer system?**
   a) Main Memory.
   b) Cache Memory.
   c) Control Unit.
   d) Central Processing Unit.
2. If we have an 8M X 64-bit memory module (Byte addressable), then the needed number of address lines is:
   a) 28.
   b) 23
   c) 26
   d) 16.
3. Which memory type requires periodic refresh?
   a) SRAM.
   b) ROM.
   c) DRAM.
   d) Both a and c.
4. The register that holds the address of the next instruction to be executed is:
   a) Status.
   b) Instruction.
   c) Program Counter.
   d) Accumulator.
5. An immediate addressing mode means that:
   a) The address of the operand is in the address stated in the instruction.
   b) The address of the operand is explicitly stated in the instruction.
   c) The address of the operand is to be found in the ALU.
   d) None of the choices.
6. In a positional numbering system, the value of a digit depends on:
   a) Total number of digits.
   b) The size of memory.
   c) Whether the number is signed.
   d) Its position and the base.
7. If a function F is equal to 1 for the expression (x+y)', then:
   a) Both inputs are 1.
   b) Only one of the inputs is 0.
   c) Only one of the inputs is 1.
   d) Both inputs are 0.
8. …… is an intelligent serial interface that supports "plug and play":
   a) Universal Serial Bus.
   b) Parallel Port.
   c) Serial port.
   d) None of the choices.
9. Which bit in a signed binary number indicates the sign?
   a) Least Significant Bit.
   b) Parity bit.
   c) Most Significant Bit.
   d) Carry bit.
10. Which memory type is non-volatile?
    a) DRAM.
    b) SRAM.
    c) ROM.
    d) Both b and c.

Part II: Short essay questions [15 marks]

This part consists of **five questions** carrying a **WEIGHT OF 3 marks for each**.
You should answer **all** questions.

Q-2: What do the following acronyms stand for: **EEPROM**, **CPU**, **IR**?

Q-3: Write the RTN of **Subt y** and **Store y**.

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Q-4: Answer the following questions:

a) List two advantages of DRAM over SRAM.
b) What does ROM store? Provide an example.

Q-5: Using MARIE ISA, find the equivalent assembly codes of the following binary instructions.

a) **1000** 1000 **0000** 0000.
b) **0110** 0000 **0000** 0000.

Q-6: Draw the characteristic table of the **S-R Flip-Flop** and **its** basic NOR-Gate representation.

Part III: Numerical problems [25 marks]

This part consists of **four questions** carrying a weight of **25 marks**.
You should **answer all the questions**.

Q-7: Suppose we have the following code: **[6 marks]**

Add 400
Load 300

Knowing that **AC=600** before executing the two instructions, and the memory and register R1 containing the values below, determine the actual value that will be found in the accumulator **AC** after executing each instruction in each mode, and fill in the table below:
Note that for each addressing mode the value of AC is initially 600

100	200
...	
200	400
...	
300	600
...	
400	300
...	
500	100
...	
600	400
...	
700	300
...	
1000	300

R1
200

Mode	Value in AC after Add 400	Value in AC after Load 300
Immediate		
Direct		
Indirect		
Indexed		

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Q-8: Answer the following for the MARIE program. Assume inputs are decimal. **[6 marks]**

a) Explain the SKIPCOND 800 decision regarding the flow of the program. (3 marks)
b) What is stored in RESULT when A = 9 and B = 12? (2 marks)
c) What will be displayed on the screen? (1 mark)

```
ORG 200
CLEAR
INPUT
STORE A
INPUT
STORE B
LOAD A
SUBT B
SKIPCOND 800
JUMP ELSE
LOAD B
SUBT A
STORE RESULT
JUMP OUT
ELSE, LOAD A
SUBT B
STORE RESULT
OUT, LOAD RESULT
OUTPUT
HALT
A, DEC 5
B, DEC 8
RESULT, DEC 0
```

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Q-9: Assume that we have a system with a memory of **256M words**. Blocks are **64 words** in length and the cache consists of **32K blocks**. **[6 marks]**

a- How many blocks are there in the main memory?

Draw the format of the memory address as seen by the cache, showing the fields as well as their sizes, for the following schemes:

b- Direct Mapped Cache.
c- 16-way set associative cache mapping.

Q-10: Answer the following questions: **[7 marks]**

a) Find the values of (-38) and (-98) in 8-bits binary representation using **C1** representation.
b) Find the sum of (-38) and (-98) in 8-bits binary representation using **C1** representation.
c) What can you say about the result? Discuss.

End of Questions